REDEFINING READING: HOW DIGITAL TOOLS ARE CHANGING FUNCTIONAL LITERACY

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This study examines how teens use digital tools like text-to-speech and generative AI to access information, revealing how these technologies both disrupt traditional definitions of literacy and push institutions to recognize multimodal reading as valid and equitable.

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INTRODUCTION

- **Functional literacy is a person's capacity to use and engage with texts to develop their potential, achieve their goals, and participate in society.**
- **Now more than ever, digital tools are an integral aspect of navigating the informational, educational, and social landscape.**

RESEARCH QUESTION

How are digital reading technologies, such as text-to-speech, speech-to-text, audiobooks, voice assistants, and generative AI, reshaping traditional definitions of functional literacy?

METHODOLOGY

Mixed-methods research approach:

- **Theoretical historical literature review**
- **Local empirical investigation**

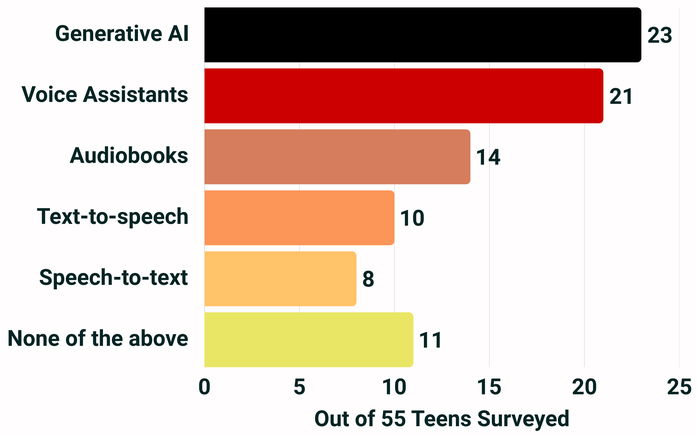
THEORETICAL FRAMEWORK

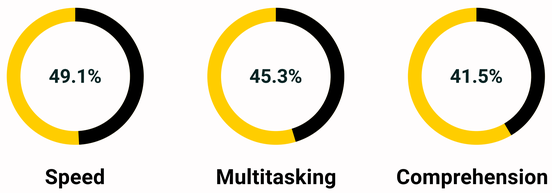
- **Universal Design for Learning (UDL)**
- **Assistive Technology**
- **Cognitive Load Theory**
- **Literacy as a Social Practice**

ANALYSIS

The Digital Disconnect

Google Form Survey About the Tools Teens Use for Reading & Writing



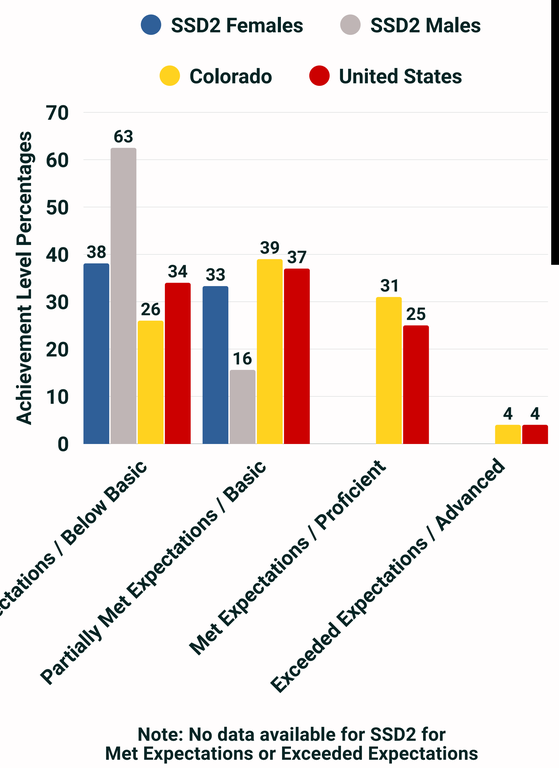


Motivations to Use Digital Tools for Reading

Data Source: Student Survey, 55 Teens Surveyed (2025)

The Literacy Gap

8th Grade Reading Test Data (2024)



Note: No data available for SSD2 for Met Expectations or Exceeded Expectations

Data Sources: Colorado Department of Education, The Nation's Report Card

RESULTS/FINDINGS

- **Text-to-speech increases comprehension for struggling readers by reducing cognitive load**
- **Multimodal digital access expands pathways to information and participation**
- **Mismatch between how teens use digital tools and literacy scores highlights opportunities for pedagogical shifts and targeted instructional interventions**

CONCLUSION

Institutional gatekeepers must redefine functional literacy by validating these multimodal digital supports as legitimate and equitable avenues to access and participation, expanding meaningful access for under-resourced youth, reducing the cognitive burdens that impede learning, and supporting fuller participation in academic and social life.

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